MIL-T-26025E(USAF)
28 DECEMBER 988
SUPERSEDING
MIL-T-0026025D(84)
31 MAY 1988

MILITARY SPECIFICATION
TRACTOR, AIRCRAFT TOWING, DIESEL ENGINE DRIVEN (DED), 4 X 4, 4 WHEEL STEER, 10,300 POUNDS DRAWBAR PULL, PNEUMATIC RUBBER TIRED

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers 4 x 4,4 wheel steer, pneumatic tired, diesel engine driven tractors intended for usc as aircraft towing and spotting vehicles for aircraft weighing up to 175,000 pounds and trailers weighing up to 93,000 pounds.

1.2 Clas sificat ion. The classification shall be as follows: Tractor, 175,000 lbs Gross Aircraft Weight.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specific ations. standards. and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of usc in improving this document should be addressed to: WR-ALC/MMVRS Robins AFB, GA 31098-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 1740

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SPECIFICATIONS

FEDERAL

W-B-131	Battery Storage: Vehicular, Ignition, Lighting and Starting
ZZ-T-381	Tire, Pneumatic, Vehicles and Portable Equipment
ZZ-I-550	Inner Tube, Pneumatic Tire
VV-F-800	Fuel Oil, Diesel

MILITARY

MIL-P-514	Plates, Identification, Instructions and Marking, Blank
MIL-T-704	Treatment and Painting of Material
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-P-23377	Primer Coating, Epoxy-Polyamide, Chemical and Solvent Resistant
MIL-C-46168	Coating, Aliphatic Polyurethane, Chemical Agent Resistant
MIL-E-52649A	Engine Cold Starting Aids, Ether Fuel Primers
MIL-C-53039	Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant
MIL-T-83133	Turbine Fuel, Aviation, Kerosene Type, Grade JP-8
MIL-H-83282	Hydraulic Fluid, Fire Resistant, Synthetic, Hydrocarbon Base, Aircraft
MIL-C-83286	Coating, Urethane Aliphatic Isocyanate, for Aerospace Applications

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-12	Abbreviations for Use of Drawings and on Specifications, Standards and Technical Documents
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Measure of Electromagnetic Interference Characteristics
MIL-STD-808	Finishes Protective and Codes for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-1223	Nontactical Wheeled Vehicle Treatment, Painting, Identification Marking and Data Plate Standards
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MIL-STD-1595	Qualification of Aircraft, Missile and Aerospace Fusion Welders
MIL-STD-1791	Designing for Internal Aerial Delivery in Fixed Wing Aircraft
MS51335-2	Pintle Assembly, Towing -18000 # Capacity, Manual Release

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COMMERCIAL ITEM DESCRIPTION

A-A- 1108	Extinguisher, Fire Vaporizing Liquid

DRAWINGS

68E34230 681334247 DL69O-1G-1O9 DL690-lG-209	Connector and Pintle Installation, MB-4 Tractor

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

LAWS A-ND REGULATIONS

49CFR 393	Federal Motor Carner Safety Regulations (FMCSR)
49CFR 571	Federal Motor Vehicle Safety Standards (FMVSS)

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, US Printing Office, Washington DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance there of.)

(Copies of specifications, standards, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 Structural Welding Code, Steel

(Application for copies should be addressed to the American Welding Society, 2501 NW Seventh Street, Miami FL 33125.)

TIRE AND RIM ASSOCIATION, INC.
YEARBOOK

DEPARTMENT OF AGRICULTURE

5100-lA Spark Arresters for Internal Combustion Engines.

(Application for copies should be addressed to Department of Agriculture Forest Service, Equipment Development Center, San Dimas, California, 91773.)

AMERICAN NATIONAL STANDARDS INSTITUTE

Square and hex bolts and screws inch series including hex cap screws and Lag screws (B 18.2.1)

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SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)
SAE Standards and Recommended Practices

J53	Minimum Performance Criteria for Emergency Steering of Wheeled Earth Moving Construction Machines
J95	Headlamps for Industrial Equipment
J293	Vehicle Grade Parking Performance Requirements
J297	Operator Controls on Industrial Equipment
J318	Trucks, Truck Tractors, and Trailers, Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers
J336a	Sound Level for Truck Cab Interior
J360	Motor Vehicle Grade Parking Performance Test Code
J366b	Exterior Sound Level for Heavy Trucks and Buses
J534	Lubrication Fittings
J680	Location and Operation of Instruments and Controls in Motor Truck Cabs
J683	Tire Chain Clearance - Truck, Buses and Combinations of Vehicles
J690	Certificates of Maximum Horse Power for Motor Trucks and Truck Tractors
J833	USA Human Physical -Dimensions
J899	Operator's Seat Dimensions - Construction and Industrial Equipment Design
J919	Sound Level Measurements at the Operator Station for Agricultural and Construction Equipment
J992B	Brake System Performance Requirements - Truck, Bus and Combination of Vehicles
J1082	Fuel Economy Measurement Road Test Procedure
ARP 1247	General Requirements for Aerospace Ground Support Equipment Motorized and Non-motorized
J1349	Engine Power Test Code - Spark Ignition and Diesel

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale PA 15096).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 Design.

3.1.1 Selection of specifications and standards. Specifications and standards, necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.1.2 Materials. Materials shall be as specified herein. Materials not specifically covered by this specification or referenced document shall be suitable in every respect for this type equipment. Except as otherwise specified, wood shall not be used.

3.1.2.1 Protective treatment. Materials that are subject to deterioration shall be protected in such a manner that shall in no way prevent compliance with the performance requirements of this specification. Protective coatings that will chip, crack or scale with age or extremes of climatic conditions shall not be used.

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3.1.2.2 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.1.3 Design and Construction The vehicle shall be constructed so that parts shall not work loose in service, and shall withstand the loads, shocks, vibrations and other conditions incident to operation, shipping, and storage with minimum loss of time for maintenance, repair, and servicing. The design of the vehicle shall ensure safe, efficient, and economical operation. All assemblies, controls, and installed equipment shall be located so there will be no adverse interference with each other or with operations and shall be readily accessible for maintenance, operation, removal, and replacement. The vehicle shall comply with MIL-STD-1472, Federal Safety Regulations 49CFR571 and 49CFR393.

3.1.3.1 Reliability. The vehicle, including all components, shall have a Mean-Time-Between-Failures (MTBF) of not less than 250 hours demonstrated at an 0.9 confidence level. The vehicle shall be designed for a life expectancy of at least 14 years.

3.1.3.2 Maintainability. The vehicle shall be designed and constructed to provide the following:

a. A minimum number of parts consistent with the performance required herein.

b. A minimum amount of time for assembling, disassembling, locating trouble sources and maintaining the vehicle. Where practical, components shall be located for rapid inspection and recognition of potential failure.

c. Ease of adjusting, servicing and replacing parts and components.

d. Use of readily available standard tools and equipment for maintenance.

e. Maintenance with a minimum number of tools.

f. Maximum repair time, excluding time to troubleshoot, of no greater than eight hours.

g. Maximum total downtime to perform all daily scheduled maintenance and servicing tasks of no greater than 15 minutes.

The vehicle shall be designed so that one mechanic, using common hand tools, can perform the following maintenance operations within the time specified:

(1) Remove, replace and adjust all engine driven belts - 3/4 hour.

(2) Remove and replace alternator - 1/2 hour.

(3) Remove and replace all filters, screens and strainers in all hydraulic systems - one hour.

(4) Remove and replace engine coolant system hoses - 1/2 hour.

(5) Drain engine lubricating oil, remove and replace oil filter element (s) and refill crankcase - one hour.

(6) Remove and replace fuel filter element (s) - 1/2 hour.

(7) Disconnect battery cables, remove and replace batteries and reconnect battery cables - 1/2 hour,

(8) Drain torque converter oil and transmission oil, remove and replace all filter elements and strainers and refill converter and transmission - one hour,

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(9) Remove and replace starter - one hour.

(10) Remove or install the engine/transmission or engine/power divider package, two mechanics - four hours.

3.1.3.3 Foolproofness. Where improper installation of an item could cause a malfunction, an asymetric mounting system shall be provided, where practical, to ensure proper mounting of the item.

3.1.3.4 Fastening devices. All screws, bolts, nuts, pins and other fastening devices shall be properly designed, manufactured and installed with adequate means of preventing loss of torque or adjustment. Cotter pins, lock washers or nylon patches shall not be used for this purpose, except for attachment of trim items on the commercial chassis. Tapped aluminum threads shall have a minimum thread engagement of at least two times the nominal fastener diameter.

3.1.3.5 Servicing provisions. Drains, grease fittings, lines and check points for all components shall be so located that they are readily accessible and do not require special tools for proper servicing.

3.1.3.6 Lubrication. All parts normally requiring grease for lubrication shall be provided with lubrication fittings in accordance with SAE J534. Where the use of high lubrication pressures may damage grease seals, a pressure relief device shall be provided. Extended fittings shall be provided to lubricate parts or assemblies which are not readily accessible for direct lubrication or which are likely to be overlooked because of inaccessibility.

3.1.3,7 Hydraulic fluid. Hydraulic fluid shall be in accordance with MIL-H-83282.

3.1.3.8 Component protection. All space in which work is performed during operation, service and maintenance of the vehicle shall be free of hazardous protrusions, sharp edges or other features which may cause injury to personnel. All rotating and reciprocating parts and all parts subject to high operational temperatures or subject to being electrically energized, that are of such nature or so located as to be hazardous to personnel, shall be guarded or insulated to the extent necessary to eliminate the hazard.

3.1.3.9 Foreign object damage. All loose metal parts, such as pins and valve caps, shall be securely attached so they cannot become separated from the vehicle.

3.1.3.10 Bonding. Metal parts shall be specifically bonded together to minimize the possibility of sparks,

3.1.4 Environments 1 conditions. The vehicle shall be capable of satisfactory storage and operation under the following environmental conditions:

a. Full-time exposure to temperatures ranging from -40°F to +125° F for Type “A" winterization and from -10° F to +125° F for Type “C” winterization (see 6.2).

b. Full-time exposure to relative humidities up to 100 percent.

c. Part-time (five percent of the time) exposure to rain; droplets of size 2.0 to 4.5 mm falling at a rate of 5.2 inches per hour.

d. Part-time (five percent of the time) exposure to salt fog; five percent solution (five parts by weight of salt in 95 parts by weight of water) with fallout of 3.0 ml/80 cm^2/hour.

e. Part-time (five percent of the time) exposure to blowing dust; velocity 1750 ft/min, concentration 0.5g/ft^3.

f. Part-time (five percent of the time) exposure to blowing sand; velocity 5700 ft/min, concentration 0.0773g/fi^3.

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3.1.5 Weight distribution. The weight distribution of the vehicle in any loading condition shall not be unbalanced or improperly distributed. The difference in load between the left and right wheels of any axle shall not exceed 2.5 percent of the total load on the axle.

3.1.6 Body lean. The vehicle shall not lean to either side more than one percent of the body width, in either the fully loaded or the unloaded condition, Body lean measurements shall be made at the outermost portion of the vehicle (excluding mirrors, mirror support brackets, lights or asymmetrical shapes) directly above each axle; the difference in body height at each axle shall not exceed one percent of the body width at that location.

3.1.7 Electromagnetic interference (EMI).

3.1.7.1 Electromagnetic interference suppression. The vehicle shall be equipped for the suppression of electromagnetic interference in accordance with Part 8 (Group I) of MIL-STD-461B as modified below. As a minimum, these bandwidths listed below shall be used in the appropriate frequency band of the measurement receiver for UM03 testing.

TUNED FREQUENCY	BANDWIDTH	TYPE BANDWIDTH
150 KHz- 1MHz	1 KHz, plus or minus 10 percent	Impulse
1 MHz -30 MHz	10 KHz, plus or minus 10 percent	Impulse
30 MHz- 1 GHz	100 KHz, plus or minus 10 percent	Impulse

The limits for UM03 (Figure 8-1) shall be relaxed by 20dB.

3.1.7.2 EMI susceptibility The vehicle shall not exhibit any malfunction, degradation of performance, or deviation from specified indications beyond the tolerances indicated in the individual equipment or subsystem specification when subjected to the following radiated electric fields (E) in accordance with MIL-STD-461B, Part 2, paragraph 19.RS03.

FREQUENCY RANGE	E-FIELD (VOLTS/METER (V/M)
2MHz to l GHz	20

3.1.8. Sound levels.

3.1.8.1 Exterior sound level. The maximum sound level produced by the vehicle shall not exceed 83dBA when measured in accordance with SAE J366b. An additional 2dBA allowance shall be permitted for production units to provide for variation in test site, temperature gradients, test equipment and inherent differences in nominally identical vehicles.

3.1.8.2 Cab interior sound level. The maximum sound level in the cab interior shall not exceed the SAE recommended design criteria for new vehicles when measured in accordance with SAE J336A.

3.1.9 Air transportability. The tractor shall be air transportable in accordance with MIL-STD-1791 in types C-130, C-141 and C-5 Aircraft. Removal of beacon light only shall be permitted, if necessary, to provide the required height for air transport.

3.1.9.1 Tiedowns. The vehicle shall be equipped with sufficient tiedown points so located that it can be restrained in all directions during air transport, The attachment points shall be designed for use with attachment in the aircraft as described in MIL-STD-1791. Sufficient clearance shall be maintained around all attachment points to allow for restraining chains without interference from any other

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structure. The tiedown points shall have a minimum clear opening of four inches to allow attachment of the chain with a grabhook.

3.1.10 Documentation and Certification. All documentation and certification shall be available during the First Article Test and upon request be furnished to the engineer during the First Article tests.

a. Required certifications from OEMs.

(1) Pintle hook rating.

(2) Axle load rating.

(3) Engine compatibility with requirements as stated in purchase description.

(4) Transmission rated for gross engine horsepower and compatibility with towing requirements as stated in purchase description.

(5) WP-ASD air transportability certification, where no park shoring is required.

(6) EMI test (with signature of appropriate QAR that was present),

(7) Miles per gallon certification.

(8) Certify gross and net horsepower and torque in accordance with SAE J690. This shall include forms shown in Figure 1 and 3. A "Description of Engine and Equipment" table and "Power Absorbing Accessories, Equipment and Attachments" table shall be included.

(9) Tire rated for 4.5 G.V.W. loadings.

(10) Cooling system will maintain the oil below transmission's recommended maximum 'effort, continuous use, temperature during 125° F ambient conditions.

b. Analytical Review. Contractor shall supply certification or verifying calculations, upon request of the Engineering Activity, that any component or system is in compliance with this specification. The calculation shall have all variables labeled and well defined, and be presented in an easy to read format.

3.1.11 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation or performance of the tractor, they shall be held or limited accordingly.

3.2 Performance.

3.2.1 Mobility. The unloaded tractor shall:

a. Maintain a forward speed of at least 15 miles per hour (MPH), plus 1.5, minus O MPH, on a level paved road.

b. Maintain a reverse speed of at least 2.5 MPH on a level paved road.

c. Ascend and descend a 12 percent grade without spillage of any fluid.

d. Negotiate a 10 percent side slope in each direction without spillage of any fluid,

e. Negotiate, in each direction, a ramp with a 20.8 percent minimum slope connecting two horizontal surfaces. The ramp length shall beat least 1.2 times the wheelbase of the

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vehicle. The vehicle shall be capable of being driven forward or backward from one horizontal surface across the ramp crest, ascending or descending the ramp, crossing the ramp crest and proceeding on the other horizontal surface without any portion of the vehicle, other than its tires, contacting any of the three surfaces.

f. Traverse Racking Course (see 4.5.11) at a speed of 8 MPH and 4 MPH.

g. Comply with all service brake performance requirements of SAE J992b.

h. Comply with all parking brake performance requirements of SAE J293, except that the required grade shall be 25 percent.

i. Have a turning radius of 180 inches maximum in either direction measured to the outside of tire tread for four wheel steer.

j. DBP (minimum at 0.75 MPH forward and reverse) 10,300 pounds.

k. Tow tractor shall be able to tow 175,000 lbs at 3 MPH, up 4 percent grade, on a surface of 0.7 coefficient of friction.

l. Tow tractor shall be able to tow a trailer up to 93,000 pounds at 15 MPH, plus or minus 2 MPH, on a level surface of 0.7 coefficient of friction.

3.2.2 Durability. The fully equipped tractor shall demonstrate a positive capability of providing operation over 2,000 miles under loads as follows:

a. One hundred miles at 10,000 pound drawbar pull.

b. Five hundred miles at 9,000 pound drawbar pull.

c. Eight hundred miles at 8,000 pound drawbar pull.

d. Four hundred miles at 6,000 pound drawbar pull.

e. Two hundred miles at 4,000 pound drawbar pull.

3.3 Chassis.

3.3.1 Powertrain.

3.3.1.1 Engine. The vehicle shall be equipped with a commercial (see 6.3.1) liquid cooled diesel engine of not less than six cylinders. The total power output required by all items driven by the engine shall not exceed its net power output rating at 2,000 feet elevation, determined in accordance with SAE J1349. The engine shall demonstrate the performance characteristics specified herein when using diesel fuel in accordance with VV-F-800 and shall operate satisfactorily on Grades JP-4 and JP-5 turbine fuel in accordance with MIL-T-5624 and Grade JP-8 in accordance with MIL-T-83133. Any engine that is adversely affected by any one of the above fuels is not acceptable. The engine manufacturer's published ratings for the engine shall not be arbitrarily raised to meet the requirement of this specification. Copy of engine manufacturer's certification to meet fuel requirements shall be provided with the end item. The engine torque, speed, cubic inch displacement and horsepower curve shall be compiled and prepared by the engine manufacturer and be submitted with the end item.

3.3.1.1.1 Engine safety devices. The engine shall have the following safety devices:

a. Low oil pressure warning light.

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b. High coolant temperature warning light (above manufacturer's recommended maximum).

3.3.1.2 Engine cooling system. The engine shall be equipped with a cooling system capable of satisfactory performance under the environmental conditions specified herein; coolant temperature shall not exceed the engine manufacturer's recommended maximum and shall be designed to 60% converter stall. Expansion tanks of sufficient capacity for expansion of all coolant in the system shall be provided. Coolant liquid (antifreeze) shall be protected to -40° F for Type C winterization and -65° F for Type A winterization.

3.3.1.3 Starting system. The starting system shall be of the electric type and shall include an ether injection system, The system shall start the engine under the following conditions:

TEMPERATURE RANGE (DEGREES F)	STARTING AIDS PERMITTED
40 through 125	None
1 through 39	Glow plugs and fluid starting aids
-40 through 0	Glow plugs, fluid starting aids and heat from the winterization system

3.3.1.3.1 Electric starter. The electric starting system shall consist of a starting motor, solenoid relay switch, neutral safety switch, starter button and/or ignition switch. The necessary wiring and cables assembled in a complete unit shall be in the form of harnesses with MS style quick disconnect.

3.3.1.3.3 Ether injection. An ether injector, in accordance with MIL-E-52649A, Type III, shall be provided. It shall be of the measured shot type and shall have thermal controls which protect the engine against ether injection after warm-up.

3.3.1.3.4 Starting system safety devices. The starting system shall have the following safety devices:

a. Shall not engage while engine is rotating.

b. Shall not crank with transmission in gear (neutral safety switch).

3.3.1.4 Exhaust system. The exhaust system shall be designed to expel exhaust gases in a manner that is not hazardous to operators or mechanics; the outlet shall be directed toward the side of the tractor. The exhaust system, including all fasteners, shall be constructed of corrosion resistant materials. Emission of sparks shall be prevented through the use of a spark arrestor or spark arresting muffler conforming to the material and performance requirements of Department of Agriculture 5100-1A.

3.3.1.5 Transmission. The vehicle shall be provided with a heavy duty automatic or semi-automatic transmission. The transmission shall transmit continuous full power through the entire gear range. The transmission shall be driven through a matched torque converter. The transmission shall permit the performance given in paragraph 3.2.1 and provide a reverse speed of not less than 2 1/2 MPH. The transmission shall be a current production model and rated for at least the maximum stall torque capacity of the torque converter, Maximum allowable transmission temperature during all operations shall not exceed 2600 F. The transmission must also be rated for gross engine horsepower and compatibility with towing requirements as cited herein (see 3.2.1A). If a hydrostatic drive system is used, a torque converter is not required,

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3.3.1.5.1 Auxiliary Transmission. A heavy duty auxiliary transmission or transfer box shall be provided to direct the power to the front and rear axles. The unit may incorporate torque multiplication. This unit shall contain an inter-axle differential of a torque proportioning type if constant 4-wheel drive is supplied, if selective 4 wheel drive is supplied no differential is required. The functions of this unit may be included in the transmission. The auxiliary transmission must meet the same rating criteria as the transmission.

3.3.1.6 Axles. The load rating of each axle shall be not less than the imposed load on the axle, measured at the ground, with the vehicle at gross weight. The axles shall be interchangeable with the changing of steering linkages and brake systems. The axle shall be equipped for steering and shall incorporate a torque proportioning differential with a bias ratio of not less than five to one or a limited slip differential. The axles shall be planetary final drive type.

3.3.2 Steering system. The steering system shall be hydraulic power boosted with a mechanical linkage (or equivalent). The vehicle shall be capable of passing the tests in SAE J53. The tractor shall furnish three modes of steering, two wheel, four wheel coordinated and four wheel oblique (crabbing). The change of mode shall be accomplished by the movement of one hand operated handle while the vehicle is in motion. The tractor shall remain in the selected mode until a new mode is selected by the operator. A capability for small increments of forward and reverse movement is required. When switching from crab or coordinated to the two wheel mode of steering, the rear wheels shall automatically center (wheel perpendicular to axle) without operator input or adjustment. Two wheeled steering mode shall function up to governed speed while turning in four wheel drive without any axle bind. The static vehicle, at idle, in four wheel steer mode, shall be capable of steering from extreme left to extreme right in a maximum of five seconds in accordance with Figure 8 of MIL-STD-1472. The steering fluid dipstick shall be accessible without removing panels and shall not subject personnel to burns or excessive reaching.

3.3.2.1 Steering wheel. The steering wheel shall have a diameter of 18 inches minimum. Not less than four inches clearance shall be provided between the steering wheel rim and the nearest projection in the plane of the steering wheel. No less than 2 1/2 inches clearance shall be provided between the steering wheel rim and the nearest projection below the steering wheel.

3,3.2.2 Electronic steer. If electronic steer is used, the following shall be met.

3.3.2.2.1 Steering lockout. In the case of any broken wheel potentiometer wire(s), the steering control system shall be locked out, leaving the rear wheels in the position they were in when the failure occurred.

3.3.2.2.2 Alarm. An alarm audible from inside and outside of the cab shall sound when the steering lockout is activated. This alarm shall be de-energized when the tractor ignition switch is turned off, but shall be re-energized should the ignition switch be turned on again prior to manually resetting the system.

3.3.2.2.3 Lockout indicator lights. A steering lockout indicator light located on the instrument panel shall be energized when the steering lockout is activated. A nameplate explaining what the operator should do in this situation shall be mounted next to the light. This Light shall be de-energized when the tractor ignition switch is turned off, but shall be re-energized should the ignition switch be turned on again prior to manually resetting the system.

3.3.2.2.4 Manual rear wheel positioning. The capability shall exist for the operator in the cab to manually position the rear wheels in the case of rear wheel steer lockout. This capability shall allow the operator to complete the current run and return the tractor for maintenance.

3.3.2.2.5 Trouble shooting lights. Trouble shooting lights, easily accessible by maintenance personnel, shall indicate whether the broken wire(s) is in the front or rear wheel potentiometer circuit. These lights shall remain energized until maintenance personnel manually reset the system. The system shall be reset by pressing a reset button or switch. The trouble shooting lights shall remain energized even with the tractor ignition in the “off’ position.

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3.3.2.2.6 Moisture protection. The potentiometer terminals and cable entry areas shall be sealed off from moisture. To reduce the effect of any moisture which might get to the potentiometers, the resistance of the potentiometers shall be 5K OHMS minimum.

3.3.3 Brakes. Stopping of aircraft while towing is totally dependent on the tractor brake system. Aircraft brakes are used only in emergency situations.

3.3.3.1 Air Brake System.

3.3.3.1.1 Service brakes. The vehicle shall be equipped with full air brakes on all wheels. Brakes shall be of either drum or disc type. Drums shall have flanges or ribbing to prevent distortion and shall be demountable without disturbing the wheel bearings. A commercial off-the-shelf air dryer and necessary associated parts shall be added to the pneumatic system to eliminate excessive moisture and oil, thus increasing the life of the air brake components. It shall use desiccant for the final absorption process and shall be installed in the discharge line between the compressor and the first reservoir. A heater and thermostat assembly shall be provided with the air dryer to preclude freeze up when used in severe winter conditions. A drain shall be provided on each air reservoir and shall be readily accessible for service. The service brakes shall be capable of stopping 175,000 pound towed load on a dry level surface (concrete or asphalt) from 5 MPH in no more than 18 feet and from 5 MPH on a three percent descending slope in no more than 33 feet. In meeting this requirement, the brakes shall not be allowed to lock up; however, it should be possible to lock up the brakes with the maximum application of the service brake pedal. The brakes shall be of the all wheel type with split circuits so that failure of one circuit shall not cause total service brake failure.

a. Physical arrangement of the tractor brake pedal and accelerator shall allow for sufficient clearance for heavy arctic gear.

b. The unloaded tractor shall not exceed 10 feet of stopping distance while traveling at 15 MPH.

3.3.3.1.2 Parking brake. The parking brake shall be capable of holding the unloaded vehicle on a 25 percent slope. Physical location of the brake control shall be within easy reach of the operator. The brake shall be of the spring applied type and shall be located on the rear wheels only.

3.3.3.1.3 Brake system safety devices. The brake system shall have low air pressure audible and visual warning devices.

3.3.3.2 Hydraulic Brake System. When specified (see 6.2) a hydraulic brake system shall be provided.

3.3.3.2.1 Brake System. The brake lines (including the hydraulic lines to the motor in a hydrostatic system) shall be securely anchored to the chassis and protected from damage. Brake lines shall not be copper or aluminum tubing. Brake lines shall be readily detachable at their axle (or motor, in the case of a hydrostatic drive system) ends. Metal protective looms shall be provided at each point where lines pass through metal members, except where a through-frame connector is provided. Connections on brake lines shall be free of 90° fittings and other restrictions that may cause delays in braking response time. Operation of the service brakes shall not result in evidence of excessive fading as governed by SAE J992B. The brake system will be functionally compatible with the towed vehicle brake system. The towing vehicle should not add to yaw, sway, skid, tip, tilt, or jackknife under maximum braking conditions.

3.3.3.2.2 Service and Parking Brakes. The service brakes shall stop the tractor within 10 feet from 15 MPH. The parking brake shall hold the tractor facing up or down a 25 percent grade. The parking brake shall be fully independent of the service brake.

3.3.3.2.3 Service Brakes. The service brakes shall be capable of stopping a 175,000 towed load on a dry level surface at 5 MPH in no more than 18 feet. The stopping distance on a three percent slope at 5 MPH shall not exceed 33 feet. In meeting this requirement, the brakes shall not be allowed to lock-up; however,

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it shall be possible to lock-up brakes with maximum application of the service brake pedal. All surfaces shall be dry, clean, rough concrete with a frictional coefficient of 0.8.

a. A maximum of a 100 lbs foot pressure shall be required to lock brakes IAW SAE ARP 1247B.

b. Braking capability shall be constant at any speed.

c. Physical arrangement of the tractor brakes shall allow brake/accelerator clearance for heavy snow boots.

d. The master cylinder shall be the dual cylinder, dual reservoir type (tandem master cylinder). Only two brakes shall have their fluid supplied per one side of the reservoir.

3.3.3.2.4 Parking Brake. The parking brake shall be capable of holding the vehicle on a 25 percent slope. Physical arrangement shall be such that at least 3 inches of clearance shall be provided for operation by an operator wearing arctic gear.

3.3.3.3 Air for Trailer Towing. When specified (see 6.2), the tractor shall comply with DL690-1G-109 and DL690-1G-209 (basic configuration including air for trailer towing), DED 7 or an equivalent system per SAE J318. When air for trailer towing is required, means shall be provided so the trailer brakes shall be activated between 0.3 and 0.6 seconds before the tractor service brakes. When air for trailer towing is required, tow additional glad hand quick connectors shall be located on the front of the vehicle and shall comply with DL960-1G- 109 and SAE J318. All glad hand quick connectors shall have covers installed on them. A minimum of 7 1/2 CFM of supplied air shall be supplied at a minimum of 110 psi and maximum of 125 psi. Air shall be modulated, proportional between foot pressure applied and air released to the trailer.

3.3.4 Frame. Frame shall be properly designed, tested, and evaluated as to ensure that the frame's load carrying capacity will be adequate to prevent any stress cracking, warping, etc. during the life of the vehicle. Frame rails shall have a minimum section modulus of 8.0 in^3 each. Frame rails and crossmembers shall have a minimum yield strength of 36,000 psi. The frame assembly shall be bolted and/or riveted; no welding shall be performed unless certification that the material has been properly normalized (stress relieved) and its strength has not been degraded is provided with First Article Report. Welded and brazed fabrication shall be in accordance with AWS D1.1 and all welders and welding equipment shall be certified by the requirements of Section 5 of AWS D1.1. No unused holes shall be permitted in the frame rails. There shall be no point on the frame that during static loading from the maximum gross vehicle weight, exceeds 2070 (safety factor of 5) of the yield strength of the frame. At no time during the operation described in paragraph 4.5.20.3 shall the frame stresses be greater than 50 percent of the material minimum yield strength (safety factor of two). Fatigue calculations from the strain gauge testing of paragraphs 4.5.20 shall project a vehicle life as follows:

a. One thousand miles over test course (see 4.5.20. 1).

b. Ten thousand maximum brake stops.

c. Ten thousand miles under loads as tested (see 4.5.20.3).

3.3.5 Suspension. A four point spring with shock absorbers (hydraulically dampened) system with a minimum of two inches of working travel must be provided. The system must provide a dynamically stable ride in both longitudinal and lateral directions. The front and rear axle frequencies shall be such that normal fore/aft pitch is minimized. The system must maintain four wheel contact with the normal working surface. MIL-STD-1472, Human Engineering Design Criteria, shall establish the limits of acceptable vibration, noise, shock and impact forces the driver is exposed to,

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3.3.6 Wheels and tires. All four tires and wheels shall be interchangeable. Tires shall be pneumatic and conform to Federal Specification ZZ-T-381. When required, tires shall be equipped with tubes in accordance with Federal Specification 22-1-550. The tires shall be of the size and ply dictated by the performance in 3.2.1 and the tractor weight in accordance with the Tire and Rim Association Yearbook. w

3.3.6.1 Tire chains. Sufficient clearance shall be provided for operation with tire chains when the vehicle is loaded to maximum vehicle weight and the tires are inflated to recommended pressure. The chains shall not strike the tractor parts. Installed chains of 5/8 inch deep cross links shall not contact any part of the tractor in any steering mode at any speed (0-15 MPH) after 5 minutes of operation to allow chains to expand. This applies to new or recapped tires of the same cross section as given in the Tire and Rim Association Handbook. This shall be in accordance with SAE J683.

3.3.7 Fuel system.

3.3.7.1 Fuel tank. The capacity of the fuel tank(s) shall ensure eight hours of normal operating duty. The tank shall not project beneath the vehicle frame and shall be adequately protected from puncture and/or structural damage. It shall be equipped with a filler tube, a safety filler cap, a float type rheostat or fuel gauge indicator, a check valve in the fuel takeoff line, a fuel shutoff cock with the fuel line connection, a drain plug at the lowest point in the tank bottom and an overflow which shall not carry fuel onto the engine muffler, or exhaust tailpipe.

3.3.7.2 Fuel filter. At least one primary filter/strainer of the fuel water separator type shall be located in the fuel line between the fuel tank and the fuel pump. At least one secondary filter for filtering fine dirt particles shall be located on the pressure (outlet) side of the fuel pump. All fuel filters shall be installed so that the elements can be readily serviced without disturbing other items. If located below the fuel tank level, a shutoff valve shall be provided between it and the tank so that the element maybe changed without draining the tank.

3.4 Cab. The body design maybe either conventional (engine in front), cab over engine, or forward control within the maximum dimensions given in 3.12. All horizontal top surfaces, including the engine cover, but excluding the cab roof, shall be capable of supporting a 200 pound man at any one foot square on the surface without permanent deformation. The cab roof shall be capable of supporting a 100 pound force at any one foot square on the surface. All normal foot traffic areas shall be skid resistant. The cab shall be provided with access panels in its floor for any top adjusted or serviced components located under the cab. All openings in the cab floor shall be sealed by boots or mats except for moisture drain holes. The cab shall conform to the requirements of MIL-STD-1472.

3.4.1 Seats. The cab shall provide two identical seats for the tractor operator and passenger, The seats shall conform in SAE J899 and MIL-STD-1472. The seats shall adjust fore, aft and vertically to accommodate an operator as large as a 95 percentile male and as small as a five percentile female in accordance with SAE J833, A full width bench seat meeting these requirements is acceptable. Seat lap belts with retractors shall be provided for both operator and passenger. Belts shall be metal to metal closure.

3.4.2 Doors. **Two cab doors shall be provided with heavy duty door limiters which hold doors in the** open position when exposed to 50 MPH winds from any direction.

3.4.3 Windshield and windows.

3.4.3.1 Windshield. The cab shall be provided with a glare proof windshield of safety plate or tempered safety glass. Two windshield wipers shall be provided. The wiper motor shall have a minimum of two speeds.

3.4.3.2 Windows. The cab shall be provided with two doors with crank regulated windows and a rear sliding window with two sliding sections.

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3.4.4 Visibility. Cab design shall provide unobstructed operator visibility forward, rearward, to either side and overhead. The cab shall be so devised as to ensure that the operator can see three inches above both the front and rear pintle hooks. If required, one mirror per pintle hook shall be provided to complete this visibility requirement.

3.4.4.1 Mirrors. An adjustable western style outside rear view mirror of not less than 50 square inches shall be mounted on each side of the vehicle on a retractable bracket. The mirrors shall be positioned so that the driver can clearly observe the roadway and vehicles approaching from the rear on either side. A spotter mirror, at lease five inches in diameter, shall be mounted below each rear view mirror. If a cab forward configuration is provided, a spotter mirror shall be mounted on each side of the windshield, installed so that the front of the vehicle is visible to the driver. An inside rear view mirror and a glare proof visor over the operator's head shall be provided. Mirrors shall, in no case, interfere with operator visibility or safety.

3.4.5 Instruments and controls. Instruments and controls shall be located within the limits specified in SAE J680, arranged so that those which are most vital to operation have the most predominant location. They shall be placed in blocks distinctly marked to group those instruments and controls pertinent to the operation of any individual component. For example, all instrument and controls relative to the operation of the engine shall be in one block. Controls having similar functions shall have similar actions. Indicator lights shall be green to indicate proper operation and red when warning is intended. All instruments and controls shall be illuminated with nonglare red lighting to be readily visible for night operation. All gauges and switches furnished on the vehicle shall be waterproof, except the ones utilized inside the cab. All gauges and switches (except spotlight switches) shall be mounted on the instrument panel. Auxiliary light switches (spotlights) shall be of a type that are easily operable by operating personnel wearing heavy winter gloves. Each gauge and switch shall be identified by a metal decal fastened to the instrument panel as to nomenclature of component, fluid being monitored, and position and function control as applicable. All instruments and controls shall conform to SAE J297. At least the following instruments and controls shall be provided:

a. Speedometer and odometer (speedometer shall read correct speed within five percent).

b. Tachometer.

c. Engine oil pressure gauge.

d. Air pressure gauge and low air warning buzzer and light.

e. Transmission oil temperature gauge.

f. Voltmeter or ammeter.

g. Fuel level gauge.

h. Brake system air pressure gauge.

i. Hourmeter.

j. Fuel control switch (engine cutoff switch).

k. Light switches.

l. Dash light (s).

m. Parking brake flasher light.

n. Low engine oil pressure warning light.

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o. Coolant temperature gauge and high engine coolant temperature warning light.

p. Low brake system air pressure warning light.

q. Rear axle centered indicating light (indicating when the rear wheels are centered).

r. Winterization receptacle power on light.

3.4.6 Climate control system. The cab's climate control system shall consist of the manufacturer's standard heavy duty heater and two outlet defroster. The heavy duty heater must, as a minimum, be capable of maintaining a 50° F' inside temperature when set at mid level during a -10° F ambient temperature for Type "C" winterization and a -40°F ambient temperature for Type "A' winterization (see 6.2). The heater shall have a three speed blower as a minimum.

3.4.6.1 Defroster fan. The tractor shall be equipped with one fully adjustable, fully enclosed fan with integral on/off switch mounted inside the cab on the cab roof. It shall be mounted approximately 12 inches from the rear window.

3.4.7 Cowl or engine cover. The engine shall be completely covered by a cowl or engine cover. On conventional design, the cowl or hood shall be hinged. The engine shall be accessible to one person.

3.4.7.1 Engine compartment. The compartment shall have sufficient access doors and access panels to permit repair maintenance and easy removal of all installed components. The engine dipstick shall be accessible without subjecting personnel to excessive reaching or bending.

3.4.7.2 cowling and access doors. All access doors other than cab doors shall be equipped with the following:

a. Door stays shall provide mechanical assistance in opening and closing doors and incorporate a positive lock to hold the doors in the open position.

b. Door latches and mechanisms shall be operable by personnel wearing arctic type gloves.

c. Door must have stops capable of stopping door when exposed to 50 MPH wind from any direction.

3.5 Electrical system. The vehicle shall be equipped with a 12 volt electrical system; a 24 volt starting system may be provided. Wiring shall be weatherproof and color coded or numbered for easy identification. All circuits shall be protected by circuit breakers located in the cab so as to be readily accessible to a mechanic. A master switch shall be provided to disconnect the batteries from the electrical system. The system shall have a negative ground.

3.5.1 Alternator. The engine shall be equipped with an alternator sufficient for operating all vehicle electrical components simultaneously. Minimum capacity of the alternator shall be 12 volts, 60 amperes, and shall have the capability to charge the battery(s) with all electrical accessories operating at engine idle RPM.

3.5.2 Batteries. The vehicle shall be equipped with at least one 12 volt battery, rated at a total of 1,800 CCA minimum. The batteries shall comply with the maintenance free characteristics of W-B-131. Batteries shall be so arranged and connected to provide sufficient capacity for engine starting in all conditions specified herein, The batteries shall be mounted in an acid-resistant tray; a sliding carriage shall be provided if the batteries must be moved sideways to permit access for servicing or replacement.

3.5.3 Lighting.

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3.5.3.1 Vehicle lighting. The vehicle shall be equipped with headlights, taillights, parking lights, clearance lights, brake lights, turn signal lights and reflectors in accordance with Federal Motor Carrier Safety Regulations. The turn signal operating units shall have visible and audible flash indicators; turn signals shall be self-cancelling. Lights and reflectors shall not be mounted on the vertical surface of rub rails or on vehicle bumpers.

3.5.3.2 Interior lighting. The cab and all compartments shall be provided with lighting adequate to enable personnel to readily obtain equipment and perform all fictions necessary during night operation. Sealed contact switches shall be provided to automatically illuminate the lights in the cab or a compartment when its door is opened.

3.5.3.3 Headlights. The tractor shall be equipped with a minimum of two automotive type sealed beam type headlights having high beam and low beam control. Lamps shall have wide dispersion type lens, shall be recessed behind the front bumper and shall be adjustable in the vertical and horizontal planes in accordance with automotive practice. Light beam shall not be obstructed by any part of the tractor. Lens shall be protected from breakage from accelerated impact (tool boxes, chock). The vehicle lights shall conform to SAE J95.

3.5.3.4 Stop and taillights. Two red reflector type combination stop lights and tail lamps of not less than 15 candle power and three candle power, respectively, shall be mounted on the rear of the tractor in a recessed or guarded location. The stop lights shall operate automatically on actuation of the service brakes.

3.3.3.5 Backup lights. The vehicle shall be equipped with backup lights mounted in a suitable rear location to prevent damage to the lights and provide light for the driver while backing. The backup lights shall be activated automatically when the tractor is placed in reverse gear. Separate switch shall provide for operation in forward gears.

3.5.3.6 Spotlight(s). The manufacturer's standard commercial spotlight (s) shall be furnished, to illuminate 180 degrees elevation can be capable of 360° rotation, forward of vehicle and to illuminate the pintle, and aircraft. towing path. Illumination of pintle, towbar, and towpath maybe provided by means of fixed lights. Controls shall be easily accessible to vehicle operator.

3.5.3.7. Reflectors and clearance lights. Reflectors and side clearance lights shall be installed on the vehicle.

3.4.3.8. Directional signals,. A directional signal switch and a hazard switch with a lighting system of two dual faced lights on the front at the sides of the grill and either the brake lights, or two dual faced lights mounted on the rear shall be provided.

3.5.3.9. Beacon. A pulsating light with amber lens shall be mounted in the center of the cab roof. The control switch shall be mounted on the instrument panel. If the beacon top exceeds the 102 inches maximum height, it must be removable from the cab roof with hand tools. The beacon light shall be the same as furnished commercially by the contractor. If the beacon has to be removed in order to meet the height requirements, means must be provided to protect the tractor horn the elements.

3.5.3.10 Pintle hook lights. Separate lights to illuminate the rear and front pintle hooks shall be provided, Switches for these lights shall be located in the instrument panel (see 3.4.5).

3.5.3.11 Domelights. Two domelights with integral individual on/off switch shall be provided. The domelights shall be mounted on the cab roof and centered over the operator's and passenger's seats (in center of adjustment range), Each light shall illuminate half of the cab adequately to allow reading of a checklist.

3.5.4 Horn. The vehicle shall be equipped with an electrically Operated horn. The operating switch shall be in the center of the steering wheel. A ring or spoke button is not acceptable. The installation of the

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button and wiring shall not admit spray nor allow water to accumulate in the wire way. The horn must be capable of being heard over the engine noise during maximum DBP operation.

3.6 Winterization system. When specified (see 6.2), the vehicle shall be equipped with a winterization system to permit storage and operation in ambient temperatures as low as -40° F. All heaters furnished shall operate on 110 volt alternating current (AC) and shall be wired through a junction block to a single three pronged (male) weatherproof slave receptacle for receiving external power and grounding the vehicle. A three w-ire connecting cable, 25 feet long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. The connecting cable shall include a matching female connector at the vehicle end and a standard weatherproof three pronged (two power, plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77 (c) (7). Electrical insulation of connecting cable shall withstand normal operating stresses in low ambient air temperature without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation and without loose excess wire. Provisions for storage of the cable shall be provided in the vehicle cab. A manual reset circuit breaker shall be installed in the circuit inside the cab to afford overload protection. A red power or indicating light shall be located on the dash panel and clearly marked. Heaters shall be furnished as follows:

a. Coolant heater, 1,500 watt minimum rating, shall be installed in the engine block or in the lower coolant inlet hose. Engine thermostat, with an operating range of 170° F to 195° F, shall be installed.

b. Immersion type engine oil heater, 300 watt minimum rating, with 170° F thermostat, shall be installed in oil pan through any convenient opening.

c. Battery heater shall be provided, having a capacity adequate to maintain the battery electrolyte at a temperature of not less than 10° F during vehicle exposure in ambient air temperatures as low as -40° F, and shall embody a thermostat to limit the temperature of the electrolyte to not more than 80° F.

d. A fuel warmer or preheater shall be provided to prevent clogging of the fuel filters due to wax crystallization in the fuel. The fuel warmer shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of -40°F to an outlet temperature of +9°F, with a flow rate of not less than the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided for the coolant inlet side of the fuel warmer unit.

3.7 Auxiliary Equipment.

3.7.1 Pintles. The hitch shall be supplied and mounted with the clevis center at a height of 14 inches above ground on both front and rear of the tractor. Pintle hooks shall have integral positive locking device and secondary positive locking device both that will engage automatically. The rear mount may be of the bolt mount design. The design load will be the rated load multiplied by a factor of three or a dynamic load multiplied by a factor of two, whichever is greater. The pintles shall be in compliance with MS 51335-2. A surge brake attachment point shall be located no more than 12 inches away horn the pintle and fastened to a structural member other than the pintle. There shall be two safety chain attaching points, one on each side of the pintle, equal distance apart, independent of pintle mounting, and capable of withstanding shock loads up to 20,000 pounds.

3.7.2 Additional Pintle and Electrical Connection for Towing MHU-85 and MHU-173E Trailers. When specified (see 6.2), the tractor shall comply with DL68E34230, or equivalent, except all electrical/receptacle requirements of the DL shall be deleted and instead, the following drawing and necessary hardware shall be included.

a. American Coleman, 540-140-009, Cable.

b. Berg, 740, Hook.

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c. American Coleman, WA-1211RROOORR, Wire.

d. Berg, 702, Receptacle (wired in accordance with SAE 506 B).

e. American Coleman, 541-88, Harness.

3.7.3. Additional pintles for aircraft towing. When specified (see 6.2, the tractor shall have an additional front and back pintle which complies with DL68E34230, except all electrical/receptacle requirements are deleted,

3.7.4 Pintle arrangment for testing. When specified (see 6.2), the tractor shall have an additional front pintle which complies with DL68E34230, except all electrical/receptacle requirements are deleted.

3.7.5 Fire Extinguishers. Two 17lb Halon 1211 Fire Extinguishers (3-A: 80B: C) in accordance with A-A-1108, Type E, shall be provided with holding brackets. They shall be mounted at the right and left side rear surface of the tractor.

3.8 Finish and protective coatings.

3.8.1 Cleaning, painting, plating, anodic films, and chemical treatments. Cleaning, painting, plating, anodic films, and chemical treatments shall be in accordance with MIL-STD-808 and/or MIL-T-704, except that commercial (see 6.3.1) components maybe prepared in accordance with the manufacturer's standard practice, provided it is compatible with the exterior finish paint.

3.8.2 Finish. The exterior paint finish shall show no evidence of uneven application, curtains, runs, sags, orange peel, lack of adhesion, or other defect and shall be capable of passing the test described in paragraph 4.8.10 of MIL-C-83286.

3.8.2.1 Forest Green. Unless otherwise specified (see 6.2), the outer surface of the vehicle and the interior surfaces of compartments (except in the cab) shall be painted with polyurethane paint, Forest Green, Color Number 24052 of FED-STD-595. The paint shall be in accordance with MIL-C-83286, *or* shall be a commercial polyurethane paint which meets the following requirements of MIL-C-83286:

(1) Paragraph 3.7.3.4 Low temperature resisitance.

(2) Paragraph 3.7.3.6.1 Accelerated weathering.

(3) Paragraph 3.7.3.7 Resistance to taping.

In addition, the paint shall have a minimum pencil hardness of 6H when tested in accordance with paragraph 4.8.12 of MIL-C-83286. All exterior trim items (such as wheels, wheel covers, running boards, metal radiator grilles, drip rails, bumpers, door handles, and light trim rings) shall be painted with the same paint as the exterior of the vehicle.

3.8.2.2 Desert Sand. When specified (see 6.2), the outer surface of the vehicle and the interior surfaces of compartments (except in the cab) shall be painted with polyurethane paint in accordance with MIL-C-46168 or MIL-C-53039, Desert Sand, Color Number 30313 of FED-STD-595. All exterior trim items (such as wheels, wheel covers, running boards, metal radiator grilles, drip rails, bumpers, door handles, and light trim rings) shall be painted with the same paint as the exterior of the vehicle.

3.8.3 Rustproofing The vehicle shall be rustproofed in accordance with the contractors' standard commercial practice provided that the requirements in 4.5.17 are met. The first article units shall not be rustproofed until after approval of the first article test report.

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3.9 Markings. Unless otherwise specified, markings for vehicles painted Forest Green shall be applied with black paint, Color Number 37038 of FED-STD-595, and markings for vehicles painted Desert Sand shall be applied with white paint, Color Number 37875 of FED-STD-595.

3.9.1. Identification markings. Unless otherwise specified (see 6.2), the vehicle shall be marked for the appropriate service in accordance with MIL-STD-1223.

3.10 Identification plates and operating instructions. All identification plates, instruction plates, data plates, etc, shall conform to MIL-P-514 and be of a size and shape consistent with the information required thereon. Abbreviations shall be in accordance with MIL-STD-12.

3.10.1 Identification date. An identification plate, permanently marked in accordance with MIL-STD-130, shall be securely attached to the vehicle in a readily accessible location. The identification plate shall contain the following information:

NOMENCLATURE

MANUFACTURER'S MODEL NUMBER

MANUFACTURER'S SERIAL NUMBER

MANUFACTURER'S NAME

MANUFACTURER'S ADDRESS

MANUFACTURER'S SERVICE PHONE NUMBER

DATE OF MANUFACTURE

CONTRACT NUMBER

NATIONAL STOCK NUMBER

3.10.2 control markings. All controls, valves, gauges and indicators used in the operation of the vehicle shall be identified by securely attached nameplates.

3.10.3 Operating instructions Brief operating and precautionary instructions shall be permanently affixed near the appropriate controls. The instructions shall be clear, concise and adequate to enable operation of the vehicle without damage to the equipment or injury to personnel and shall refer to the components as identified by nameplates.

3.10.4 Lubrication plate. A lubrication plate shall be provided directing attention to all lubrication fittings and components which require lubrication. The plate shall identify the type and grade of lubricant required for all operational temperatures.

3.10.5 Transportation data plate. A transportation date plate shall be provided. The plate shall contain at least the following information:

a. Side and rear silhouette views of the vehicle.

b. Horizontal and vertical location of the center-of-gravity of the vehicle in transport condition.

c. Shipping weight.

d. Loading cubage.

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e. Overall length, width and height.

f. Tiedown instructions for air transport in Types C-130, C-141 and C-5 Aircraft.

3.11 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.12 Dimensions. The vehicle, in fully operational conditions, including all equipment and accessories, shall not exceed the following dimensions:

a. Length (maximum) -191 inches.

b. Width (maximum) -98 inches.

c. Height (maximum) -102 inches.

d. Ground clearance (minimum) - 10 inches.

3.13 Workmanship. The vehicle, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs and sharp corners and edges; accuracy of dimensions, surface finish and radii of fillets; thoroughness of soldering, welding, brazing, painting, wiring and riveting, marking of parts and assemblies; alignment of parts and tightness of assembly fasteners, etc.

3.13.1 Welders and welding.

3.13.1.1 Welders. All welders employed in the fabrication of the vehicle, except for the commercial chassis, shall be certified in accordance with the applicable requirements of MIL-STD-1595, the American Welding Society or the American Society of Mechanical Engineers.

3.13.1.2 Welding. The surface parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. Welds shall transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test, proof and service loadings.

3.13.2 Screw assemblies. Unless torque is specified, treaded fasteners shall be tight and shall not work loose during testing or service usage.

3.13.3 Riviting. Riveting operations shall be performed to ensure tight fit and complete heading of the rivets.

3.13.4 Gear and lever assemblies. Gear and lever assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots or other irregularities. Where required for accurate adjustment, gear assemblies shall be free of backlash.

3.13.5 Cleaning. The vehicle shall be thoroughly cleaned. Loose, spattered or excess solder, welding slag, stray bolts, nuts and washers, rust metal particles, pipe compound and other foreign matter shall be removed during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contractor purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of Tests. The inspection and testing of the vehicle shall be classified as follows:

a. First Article tests (4.3.1 through 4.4)

b. Acceptance tests (4.3.1, 4.3.3 and 4.3.4).

4.3 First article tests.

4.3.1 First article test samples. The first article test samples may be preproduction items which meet the requirements of this specification. The samples shall be identical to the items procured under the terms of the contract. Approval of the first article samples shall not constitute a waiver by the Government of its rights under the provisions of the contract. Two first article samples shall be produced and tested as follows:

a. Contractor test sample (see 4.3.2.1).

b. User test sample (see 4.3.2.2),

4.3.2 First article tests.

4.3.2.1 Contractor test sample. The contractor test sample shall be subjected to the tests described in 4.5. All tests with the exception of 4.5.14 shall be run using JP-8 fuel conforming to MIL-T-83133.

4.3.2.2 Usertest sample. The user test sample shall be subjected to the individual testing (see 4.5) prior to the test described in 4.6.

4.3.3 Notification of tests. The procuring activity shall be notified of the time and location of all first article tests so that representatives of the procuring activity may attend. The notification shall be in writing and shall include the nomenclature and the national stock number (NSN).

4*3.4 Test report. Upon completion of the first article tests, a first article test report shall be prepared in accordance with MIL-STD-831 An authorized Government representative shall countersign all data sheets and the report. The following shall be incorporated as a part of the test report:

a. All failures, maintenance performed and other events recorded shall be identified by accumulated operation time, miles, cycles or position of the test procedure, as appropriate. Test conditions at the time of failures or irregular operations identified shall be recorded.

b. Summary of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.

c. Summary of the engineering reasoning behind any corrections made, to be made or proposed to be made, and the predicted effectiveness of those corrections.

d. List of all cleaning and finishing procedures to be used in accordance with MIL-STD-808 and/or MIL-T-704.

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List of methods to be used to protect against electrolytic corrosion of dissimilar metals in intimate contact in accordance with MIL-STD-889.

f. Test report for each component covered by a document listed in the DODISS.

4.4 Acceptance testing. Acceptance testing shall be classified as follows:

a. Individual testing (see 4.4.1).

b. Random sample testing (see 4.4.2).

4.4.1 Individual testing. Each vehicle shall be subjected to the tests specified in 4.5.1, 4.5.2 and 4.5.3.

4.4.2 Random Sample Test. Any time chosen by the Air Force during the production of vehicles to meet the total contractual requirement, a production line vehicle may be identified for testing. This test shall incorporate all tests required in Section 4.5, except 4.5.8, 4.5.9, and Air Transportability Certification, and be in accordance with the previously approved test procedures.

4.4.2.1 Rejection and retest. When a vehicle fails to meet the requirements of this purchase description, vehicles to be delivered pursuant to the contract may not be accepted until the cause of the failure is determined, corrective action is taken by the contractor, and the part that failed is successfully retested to assure compliance with this specification. Tests not involving the failed part may be continued pending the investigation of a test failure,

4.5 Test methods.

4.5.1 Examination of product. The vehicle shall be inspected to determine compliance with the requirements specified herein. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and their application, welding and riveting, and markings. For the first article and random sample vehicles, this examination shall be accomplished by use of a checklist which lists each requirement not validated by a test and the results of the examination.

4.5.2 Lubrication Inspection. Observe the lubrication of every fitting on the vehicle to determine compliance with:

a. Lubrication by standard military lubricants.

b. All fittings accessible.

c. Fittings is vulnerable positions are protected by recessing or bossing.

4.5.3 Draw Bar Pull. The vehicle shall be tested to show conformance with 3.2.1j. The test shall only be conducted in an environment where the dry barometer pressure is between 28 to 30 in. Hg and ambient air temperature is between 60 to 110° F. Subsequent tractors shall be tested for static drawbar pull of 11,000 pounds minimum in forward and reverse gears. In each case, diesel fuel conforming to MIL-T-83133 must be used.

4.5.4 Torque Converter Test. Immediately following the test in 4.5.3, the tractor shall be stopped by application of the service brakes and held stationary by the brakes while the transmission is in the lowest gear. The engine shall be run at 90% of torque converter stall speed for one minute. The transmission oil temperature shall not exceed 230°.

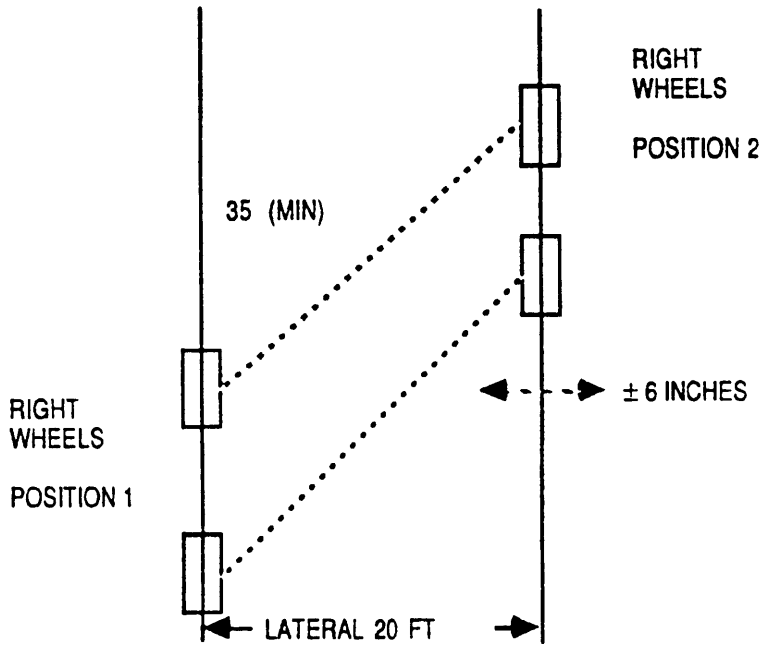
4.5.5 Endurance Test. The tractor shall be driven through continuous sharp “s” curves in four wheel drive coordinated steer for a distance of 5 miles on the odometer, and shall be accelerated through gears to 15 MPH, stopped, and backed 15 feet, in cycles for 8 additional miles on the odometer. This test shall be repeated for four wheel drive, two wheel steer.

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4.5.6 Steering Test

4.5.6.1 Turning test. At the end of the test in 4.5.5, the tractor shall be driven in full 360 degree turns to the right and left, and the turning radius shall not exceed 180 inches measured to outside of the tire tread (per 3.2.1i.)

4.5.6.2 Crab steering. The tractor shall be placed in crab steering mode and the front wheels turned at a 35° angle (minimum). The vehicle shall be driven a lateral distance of 20 feet measured from the original front wheel position. The lateral travel of the rear wheel shall be equal to 20 feet ± 6 inches.



4.5.7 Brake Test. The tractor shall be capable of stopping itself in 10 feet when traveling 15 MPH. A fifth wheel shall be used and stopping distance accomplished shall not exceed requirement from any one of six tires. Compliance with parking brake requirements shall be demonstrated. When air is required for trailer operation, delayed application of tractor brakes shall be demonstrated. Additionally, the vehicle shall be shown, by mathematical calculation, to conform with all other requirements in paragraph 3.3.3.1.1. The brake test shall be run after the endurance test (see 4.5.5).

4.5.7.1 Parking brake tests. The vehicle shall be tested in accordance with SAE J360 to demonstrate compliance with 3.2.1 (h).

4.5.7.2 Brake Test at Curb Idle. The vehicle shall be tested as follows:

Apply the service brakes, pedal force not to exceed 100 pounds, with the engine operating at curb idle, move the transmission from neutral to forward drive, back through neutral to reverse. There shall be no movement of the tractor in either direction due to wheel rotation because of brake slippage.

The parking brake shall not be applied during this test. This test shall be repeated ten times in each direction. The entire ten repetitions of the test shall be repeated using the parking brake. The service brakes shall not be applied during this portion of the test, and the parking brake shall have been set for at least eight hours prior to the beginning of the test.

4.5.8 EMI Test. The vehicle shall conform to the requirements of paragraph 3.1.7 when tested in accordance with the procedure of MIL-STD-462, Notice 2.

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4.5.9 Fuel Consumption. The vehicle shall be tested for fuel consumption at 5 MPH exerting 2,000 DBP and at 15 MPH with no load. Each speed will be tested for one hour. The distance shall be verified by fifth wheel and test track shall be relatively level. The fuel measuring device and all fuel economy correction shall be in accordance with SAE J1082.

4.5.10 Tire Chains. Type (RS) tire chains shall be installed on all four wheels and the tractor driven in each of the three steering modes. After 5 minutes of driving it shall be shown that sufficient clearance exists.

4.5.11 Racking Test. The tractor shall be driven over a course of 12 tractor wheel base length (WBL). One wheel track shall have timbers spaced at 0.3 WBL. The outer track shall have timbers shaped at 0.4 WBL. 4 X 4 timbers shall be used. Timber spacing shall be fixed by rope or cable connection. The test shall consist often round trips over the course each at speeds of 8 MPH and 4 MPH (79 timbers will be required).

4.5.12 Human Factors Engineering (HFE) Evaluation Two operators (8 inches difference in height) shall demonstrate that all controls, switches and any other components are located so as to assure operators can reach, remove, manipulate, maintain, or operate them (see 3.1.10) Per MIL-STD-1472.

4.5.13 Sound Level Test. The sound level inside the cab at the operator's ear level and outside the cab shall be measured and recorded. Tractor shall meet requirements of 3.1.8.

4.5.14 Alternate Fuel Test. The fuel tank shall be drained. The vehicle shall use JP-4 as fuel and drive 90 miles at maximum attainable speed. At the end of this test the vehicle shall show it can obtain 8,200 lbs DBP while using JP-4.

4.5.15 Emergency Steering Test. The emergency steer system specified in 3.3.2 must be run tested and conform to SAE J53.

4.5.16 Axle Bind Test The tractor must be placed in two wheel drive, two wheel steer configuration and be driven 360 degrees through the tightest turning radius obtainable. The number of rotations each wheel makes must be counted and recorded. This must be repeated in the four wheel drive, two wheel steer configuration. The number of revolutions each wheel makes in four wheel drive must be within five percent of the number of revolutions that same wheel made in two wheel drive. Immediately following the above cited test, the vehicle shall be placed in four wheel steer mode and the turning radius measured. It shall be within the limits specified (see 3.2. ii).

4.5.17 Corrosion Test Three panels, three foot by two foot, rustproofed in accordance with the contractor's standard commercial practice and painted in accordance with 3.8.2.1 of this specification, shall withstand 120 hours of a five percent concentration salt spray in a closed environment without any evidence of corrosion.

4.5.18 Cold Start Test. The vehicle shall be started in an ambient temperature of -40°F. Ether injection, as well as specified heaters stated in 3.6, shall be the only starting aids used. The cold soak period shall be a minimum of 24 hours. The vehicle shall then be cold soaked for 24 hours at -10°F, and ether injection shall be the only starting aid used.

4.5.19 Strain Gauge testing A strain gauge test shall be performed on the tow tractor frame.

4.5.19.1 Test course. The tractor shall be driven over a course of 12 tractors wheel base length (VVBL). One wheel track shall have timbers spaced at 0.3 WBL. The other track shall have timbers spaced at 0.4 WBL. Four by four (rough cut) timbers shall be used. Timber spacing shall be fixed by rope or cable connection. The test shall consist of four round trips over the course at a speed of 8 MPH and 4 MPH. All materials required for test course will be furnished by the contractor.

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4.5.19.2 Patterning. High stress areas of the frame assembly shall be coated with brittle lacquer. The tractor shall be driven over the test course at a speed of 8 MPH and 4 MPH, it shall perform a maximum stop from 15 MPH, and perform a drawbar pull of 10,000 miles for 1/2 mile.

4.5.19.3 Strain recording Strain gauge rosettm shall be positioned in locations of high stress concentrations. The locations should be determined as indicated by the brittle lacquer patterning and a minimum of 30 strain gauges shall be used. Strains shall be recorded for each of the three following tests:

a. Four round trips over the test course.

b. Five maximum stops from 15 MPH.

c. Twenty miles under load as follows:

(1) One mile at 10,000 pound drawbar pull.

(2) Five miles at 9,000 pound drawbar pull.

(3) Eight miles at 8,000 pound drawbar pull.

(4) Four miles at 6,000 pound drawbar pull,

(5) Two miles at 4,000 pound drawbar pull.

Strain/occurrence histograms shall be supplied for each gauge for each test cycle. The actual test course shall be recorded on the histogram. A picture of each gauge indicating its position shall be provided.

4.5.19.4 Evaluation. The minimum ultimate tensile and yield strengths of the gauge components shall be documented. Any strain recorded indicating stresses greater than 50 percent of the material minimum yield strength shall be cause for rejection. Fatigue calculations shall project a vehicle life as follows: _

a. One thousand miles over test course.

b. Ten thousand maximum brake stops.

c. Ten thousand miles under loads as tested.

4.5.20 Reliabilitv/durability testing Compliance with the quantitative reliability requirements of 3.1.3.1 shall be demonstrated by the following tests. Apparent failures that are caused by some unintended factor external to the equipment (e.g. test equipment failure) shall not be counted as reliability failures if the cause is substantiated to the satisfaction of the procuring activity. Any test resulting in a failure shall be fully documented. Servicing, preventive maintenance, or adjustments authorized by this specification or the procuring activity shall not be counted as failures.

4.5.20.1 Vehicle durability test. The tractor shall be driven 2,000 miles under load as follows. During the tests, tractor shall be stopped by the service brakes at least once in each five miles.

a. One hundred miles at 10,000 pounds drawbar pull.

b. Five hundred miles at 9,000 pounds drawbar pull.

c. Eight hundred miles at 8,000 pounds drawbar pull.

d. Four hundred miles at 6,000 pounds drawbar pull.

e. Two hundred miles at 4,000 pounds drawbar pull.

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4.5.20.2 Maintainability demonstration. Compliance with the maintenance downtime requirements shall be demonstrated by task performance. All scheduled maintenance tasks and all tasks listed in 3.1.3.2 shall be performed. In addition, corrective maintenance tasks shall be selected by the procuring activity from an approved list of tasks provided by the contractor in the first article test procedure. Not less than 25 percent or more than 50 percent of the total number of tasks shall be demonstrated to verify an average maintenance downtime within the required limit. As part of this demonstration, the recommended frequencies of the scheduled maintenance tasks and the times recorded to accomplish the tasks shall be used to develop an expected value of scheduled maintenance downtime per measure of use, such as calendar time, hours of operation or miles traveled. The capability of performing maintenance tasks be personnel wearing arctic mittens shall be demonstrated.

4.6 User Test. If specified (see 6.2) the vehicle shall be delivered to the Government site (see 6.2) for 60 available days (seven days a week) of user test. The 45 available days shall be consecutive, except when interrupted for maintenance/repair downtime, as defined herein. The user test will not begin until after the contractor has completed the first article test on the contractor test sample as defined by 4.5, except alternate fuel test (see 4.5.14), corrosion test (see 4.5.17) and cold start test (see 4.5.18). Acceptance tests (see 4.4) and the examination of product (see 4.5.1) shall be perfomed on the user test sample prior to shipment of the user test sample. The user test sample shall not be shipped without prior 15 day notification to the PCO. The test period shall begin after the vehicle has been inspected and prepared for towing aircraft/trailers and shall end when 45 days of availability are complete or 60 calendar days have elapsed, whichever occurs first. If the user test sample has not been available for 45 test days after 60 calendar days have elapsed, the user test shall be considered a failure. The vehicle shall be available for any one period of eight consecutive hours during each day of the 45 day test. The periods shall be scheduled by the site's representative, and the schedules shall be provided to the contractor at least 12 hours in advance. Any period of more than four hours of the same day that the vehicle is out of service for maintenance or repair shall not be considered an available day for the test period. The vehicle shall be operated, and scheduled maintenance shall be per-formed by Air Force personnel using the operation and maintenance procedures contained in the preliminary technical data to be supplied with the test sample. The preliminary technical data shall contain procedures for preparation/tiedown of the tractor for air shipment, The contractor will provide any special tools/equipment required to perform operators maintenance/scheduled maintenance. any parts or corrective actions required shall be provided by the contractor. Any corrective actions performed by the contractor shall not begin before notification to the site's representative, Government personnel shall be allowed to observe their actions. The contractor shall provide all tractor consumables required in the course of the test except for diesel fuel, water, compressed air and electrical power which will be furnished. The contractor shall provide on-site familiarization to the operators and mechanics as necessary to perform this test. Test operations will be used in or simulate normal Air Force usage and will not exceed any performance requirements defined by Section 3 of this purchase description. Noncompliance with the performance requirements of this purchase description shall be cause for rejection. The contractor shall maintain a daily log. The log shall be signed by the location's representative and the contractor after each day of activity. The daily log shall document, but not be limited to, the following:

a. Operating time.

b. Tasks completed.

c. Technical manual errors.

d. All failures and corrective actions taken.

e. Operational problems encountered.

f. User test representative comments.

The contractor shall report the user test results, including daily logs of activities, in the first article inspection report.

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4.7 Finite element analysis verification. It shall be shown by the use of finite element analysis, which shall include a working paper, stress plots and node analysis, that the frame proposed by the manufacturer for the fabrication of this vehicle shall maintain its structural integrity over the life of the vehicle. The analysis shall include all stress, dynamic loading, vibrational distortions and shock loading that would be imposed on the frame during normal operational use and in accordance with the performance specification described in this specification. Normal use shall include, as a minimum, an unloaded maximum stop and a maximum drawbar pull. The frame materials allowable stress used in this frame design shall be in accordance with the following order of precedence:

a. SAE Standards.

b. ASTM Standards.

c. ASME Standards.

d. AISC Standards.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with the contract.

6. NOTES

6.1 Intended Use. The tractors furnished under this document will be used for towing aircraft whose gross weights are not more than 175,000 lbs and trollers whose gross weights are not more than 93,000 lbs, under existing weather and grade conditions.

6.2 Ordering Data. Procurement documents should specify the following:

a. Title, number, and date of this document.

b. Type of winterization required (see 3.1.4,3.4.6, 3.6).

c. Hydraulic Brakes required (see 3.3.3.2)

d. Air for tow trailer (see 3.3.3.3).

e. Additional pintle and electrical connection (see 3.7.2).

f. Finish color required, if not Forest Green, Color Number 24052 of FED-STD-595 (see 3.8.2.1, 3.8.2.2).

g. Additional pintle for aircraft towing (see 3.7.3).

h. Pintle arrangements for testing (see 3.7.4).

i. Identification required if other than MIL-STD-1223 (see 3.9.1).

j. If user testis required and government site of user test (see 4.4).

6.3 Definitions.

6.3.1 Commercial. A manufacturer's standard product, having been marketed, in production and sold in substantial quantities to the general public for a minimum of one year prior to the date of the invitation for bid or request for proposal. An item is sold in substantial quantities only when the quantities regularly sold are sufficient to constitute a real commercial market. Nominal quantities,

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such as models, samples, prototypes or experimental units, do not meet this requirement. The general public is a significant number of buyers other than the Government or afilliates of the offeror; the item involved must not be for Government end use. For the purpose of this specification, items acquired for ("Government end use") include items acquired for foreign military sales.

6.4 Subject Term (Key Word) listing.

Aircraft Towing

Diesel Engine Driven (DED)

Pneumatic Rubber Tired

Tractor

6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force -99

Preparing activity:
Air Force -84

Project No: 1740-F163

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-26025E(USAF)	

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a Paragraph Number and Wording

b Recommended Wording

c Reason/Rationale for Recommendation

6. REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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